16. The truth table for a full adder is shown below. What are the values of X, Y, and Z?

A	B	Cin	Cout	∑
0	0	0	0	0
0	0	1	0	**X**
0	1	0	0	1
0	1	1	**Y**	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	**Z**

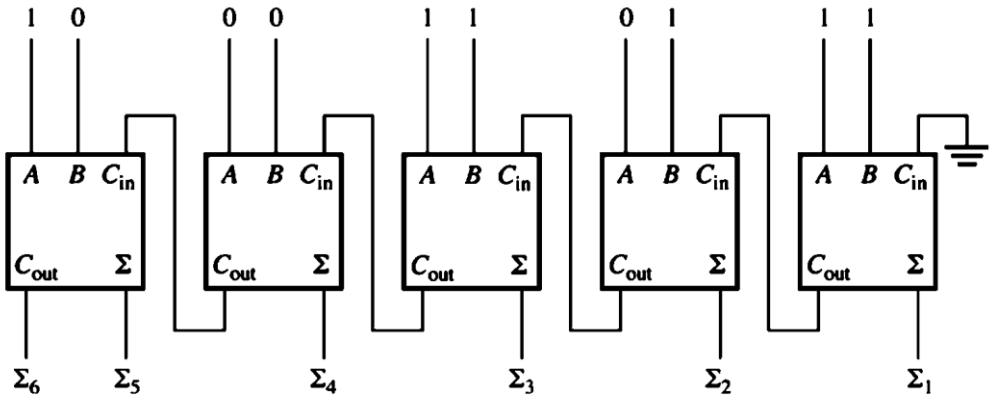
A. X = 0, Y = 1, Z = 1

B. X = 1, Y = 1, Z = 1

C. X = 1, Y = 0, Z = 1

D. X = 0, Y = 0, Z = 1

17. What is the correct result for the following parallel adder?



A. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 11100

B. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 00111

C. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 01110

D. $\sum_6\sum_5\sum_4\sum_3\sum_2\sum_1$ = 11110

18. The binary numbers A = 1100 and B = 1001 are applied to the inputs of a comparator. What are the output levels?

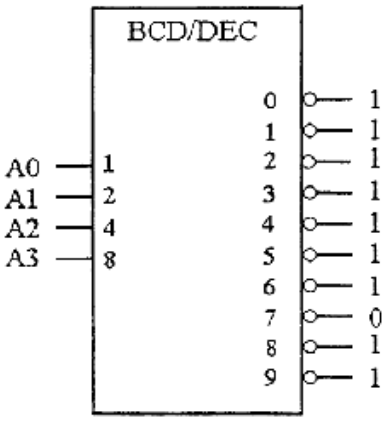
    A. A > B = 1, A < B = 0, A < B = 1

    B. A > B = 0, A < B = 1, A = B = 0

    C. A > B = 1, A < B = 0, A = B = 0

    D. A > B = 0, A < B = 1, A = B = 1

19. What type of device is shown below, and what inputs (A3, A2, A1, A0) are required to produce the output levels as shown?



    A. BCD-to-decimal encoder; 0,1,1,1

    B. BCD-to-decimal encoder; 1,1,1,0

    C. BCD-to-decimal decoder; 1,1,1,0

    D. BCD-to-decimal decoder; 0,1,1,1

20. A priority encoder means that _____.

    A. the lowest priority goes first.

    B. the highest input has priority.

    C. priority is programmed into the device.

    D. the lowest input has priority.

QUESTION 4 (12 POINTS)

Figure 6 shows 4 bits data router using single MUX and DEMUX connection. Inputs MUX are labeled as X_3:X_0 while outputs DEMUX are Y_3:Y_0. Bit select for MUX and DEMUX are noted by C_1 and C_0 (C_1 is the Most Significant Bit (MSB)). *A* is noted as the output of MUX and *B* is the input of DEMUX.

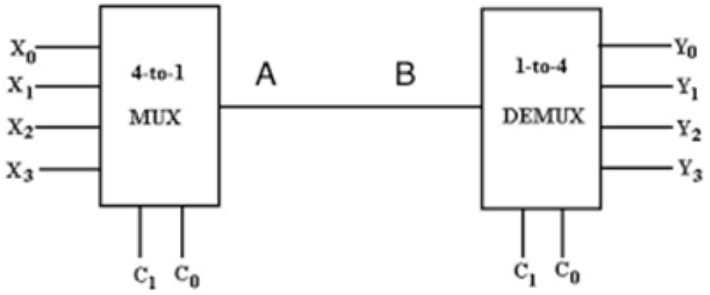


Figure 6

A. Complete the truth table for MUX and DEMUX. List all possible values. [4 pts]

Truth Table: MUX (4 to 1)

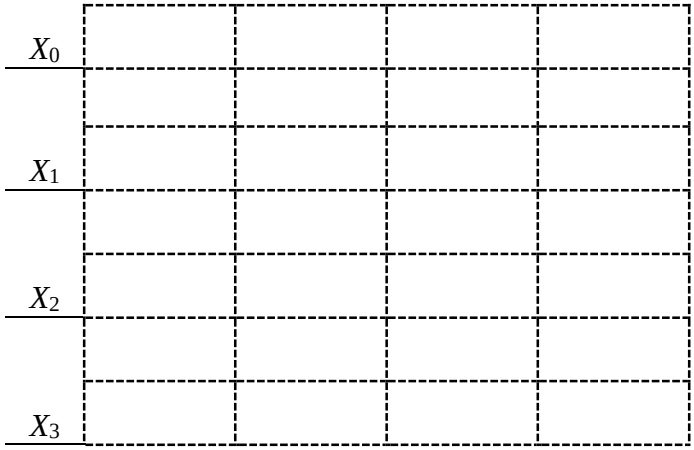
C_1	C_0	A

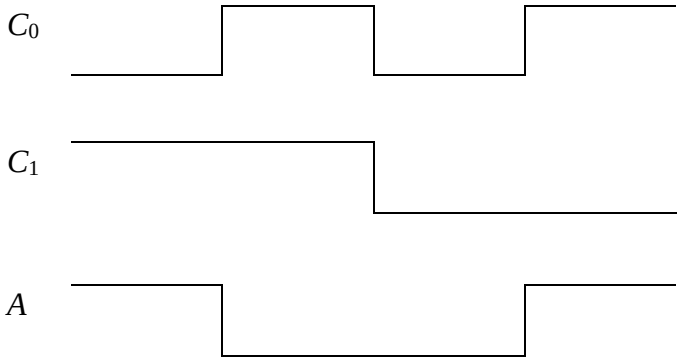
Compact Truth Table: DEMUX (1 to 4)

C_1	C_0	Y_3	Y_2	Y_1	Y_0

B. Analyze the following MUX operation. Draw the possible inputs for X_0 until X_3 based on the given Y output. [4 pts]

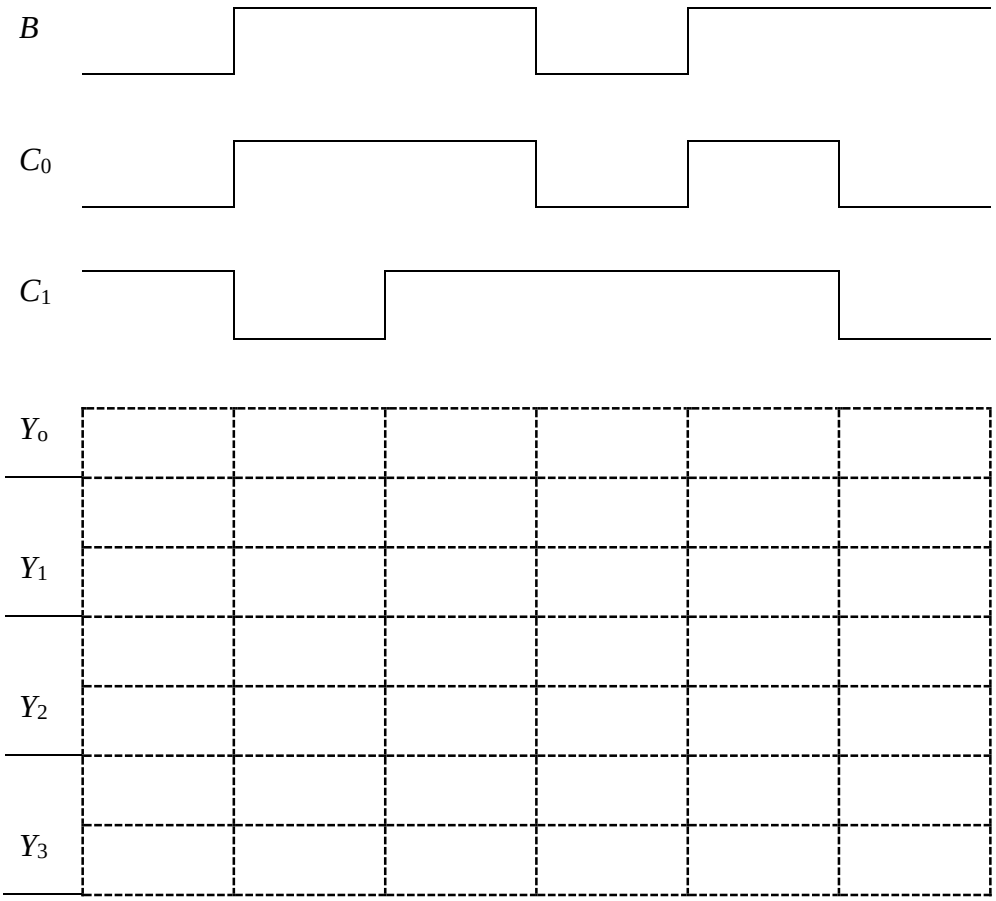
MUX operation





C. Analyze the following DEMUX operation. Draw the timing diagram for the required DEMUX output based on the given values. [4 pts]

DEMUX operation



QUESTION 5 (20 POINTS)

Figure 7 shows a fruit basket that will be used to lift a bunch of apples to the main container. The fruit basket can contain a maximum of 4 apples. Initially the fruit basket is empty. We have a bunch of apples that we want to put into the basket one by one. Once the basket is full, we have to lift the basket to the main container and pull out the apples one by one until the basket is empty. Sensor X will sense the counting sequence of the counter while sensor Z controls the movement of the lid.